

Quantifying the impact of proactive community case management on severe malaria cases using agent-based simulation

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**Health and
Humanitarian Systems**

Malaria



What is Malaria?

- Caused by a parasite transmitted through Anopheles mosquito bites
- ~ two-week incubation time



Common Symptoms

- Fever, chills, headache, muscle aches, often as flu-like illness
- Progress to severe malaria shortly, especially for children



Risk Worldwide

- 2022: 249M cases, 608K deaths, mostly in Sub-Saharan Africa
- Very low risk in USA, mostly imported



Prevention & Treatment

- Mosquito nets
- Indoor spraying
- Test and Artemisinin-based Combination Therapy (ACT)

ProCCM



What is ProCCM?

- Proactive Community Case Management, regular home visits by health workers in entire community



What can ProCCM do?

- Identify suspected malaria cases, provide testing, and administer treatment



Why ProCCM?

- Enhance early case detection and access to treatment
- Increase healthcare worker utilization



Additional note

- Sweep: one round of home visits
- Routine treatment seeking for villagers continues as usual

Literature

Pilot Study Background [1]



July-Nov 2013 (20 Weeks), Saraya Department, Kedougou, Senegal



Lower symptomatic malaria prevalence in intervention villages



Previous study of the impact of ProCCM on symptomatic cases [2]



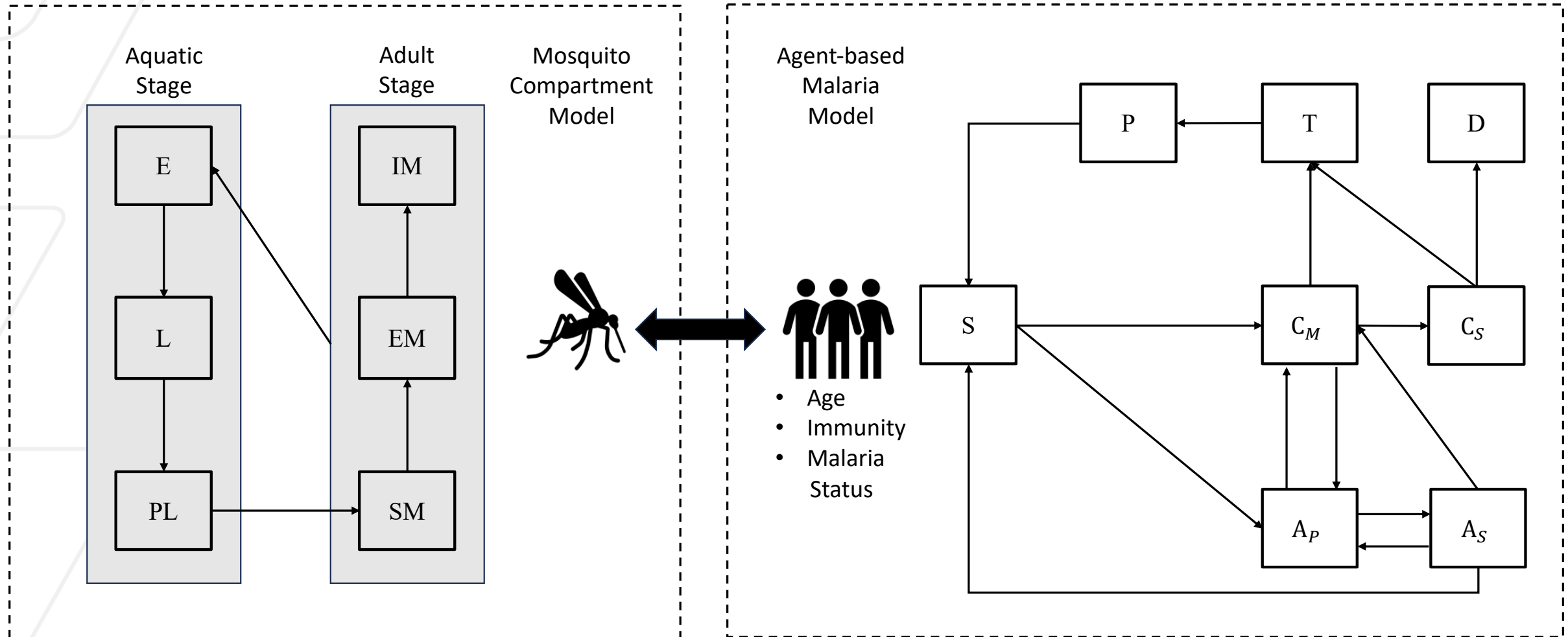
Research Question

1. Better strategies for **allocating sweeps** to reduce severe cases
2. Impact of the number of sweeps on **severe Malaria cases**
3. Impact of **treatment seeking behavior** on the effectiveness of ProCCM

[1] Linn, A. M., Ndiaye, Y., Hennessee, I., Gaye, S., Linn, P., Nordstrom, K., et al. 2015. "Reduction in Symptomatic Malaria Prevalence through Proactive Community Treatment in Rural Senegal". Tropical Medicine & International Health 20(11): 1438-1446.

[2] Wang, Y., Wang, X., Gurbaxani, B., Gutman, J. R., Keskinocak, P., Smalley, H. K., et al. 2024. "Modeling the Impact of Proactive Community Case Management on Reducing Confirmed Malaria Cases in Sub-Saharan African Countries". The American Journal of Tropical Medicine and Hygiene 111(3): 490.

Transmission model



[3] Griffin, J. T., Bhatt, S., Sinka, M. E., Gething, P. W., Lynch, M., Patouillard, E., *et al.* 2016. "Potential for Reduction of Burden and Local Elimination of Malaria by Reducing Plasmodium Falciparum Malaria Transmission: a Mathematical Modelling Study". *The Lancet Infectious Diseases* 16(4): 465-472.

[4] White, M. T., Griffin, J. T., Churcher, T. S., Ferguson, N. M., Basáñez, M. G., Ghani, A. C. 2011. "Modelling the Impact of Vector Control Interventions on Anopheles Gambiae Population Dynamics". *Parasites & Vectors* 4: 1-14.

Transmission model



Mosquito Compartment Model

- Transition captured by sets of differential equations (updated daily)
- **Aquatic Stage** (Biological)
 - E (Egg) $\rightarrow$ L (Larva) $\rightarrow$ PL (Pupa/Late stage)
- **Adult Stage** (Infectious)
 - SM (Susceptible) $\rightarrow$ EM (Exposed) $\rightarrow$ IM (Infectious)



Agent-Based Human Model

- Each person (agent) modeled with:
 - **Age** (in days)
 - **Immunity** (four types with different role)
 - **Current Malaria Status**
- Stages consist of:
 - C_M/C_S : mild/severe malaria
 - A_P/A_S : patent/subpatent asymptomatic
 - S, T, P, D : susceptible, treatment, protection and death

TSR and Sweeps



Treatment Seeking Rate (TSR)

- Represents routine care behavior in the absence of interventions
- **Daily probability** of a symptomatic patient seeking care
- Applied uniformly across the population

[1] Linn, A. M., Ndiaye, Y., Hennessee, I., Gaye, S., Linn, P., Nordstrom, K., et al. 2015. "Reduction in Symptomatic Malaria Prevalence through Proactive Community Treatment in Rural Senegal". Tropical Medicine & International Health 20(11): 1438-1446.



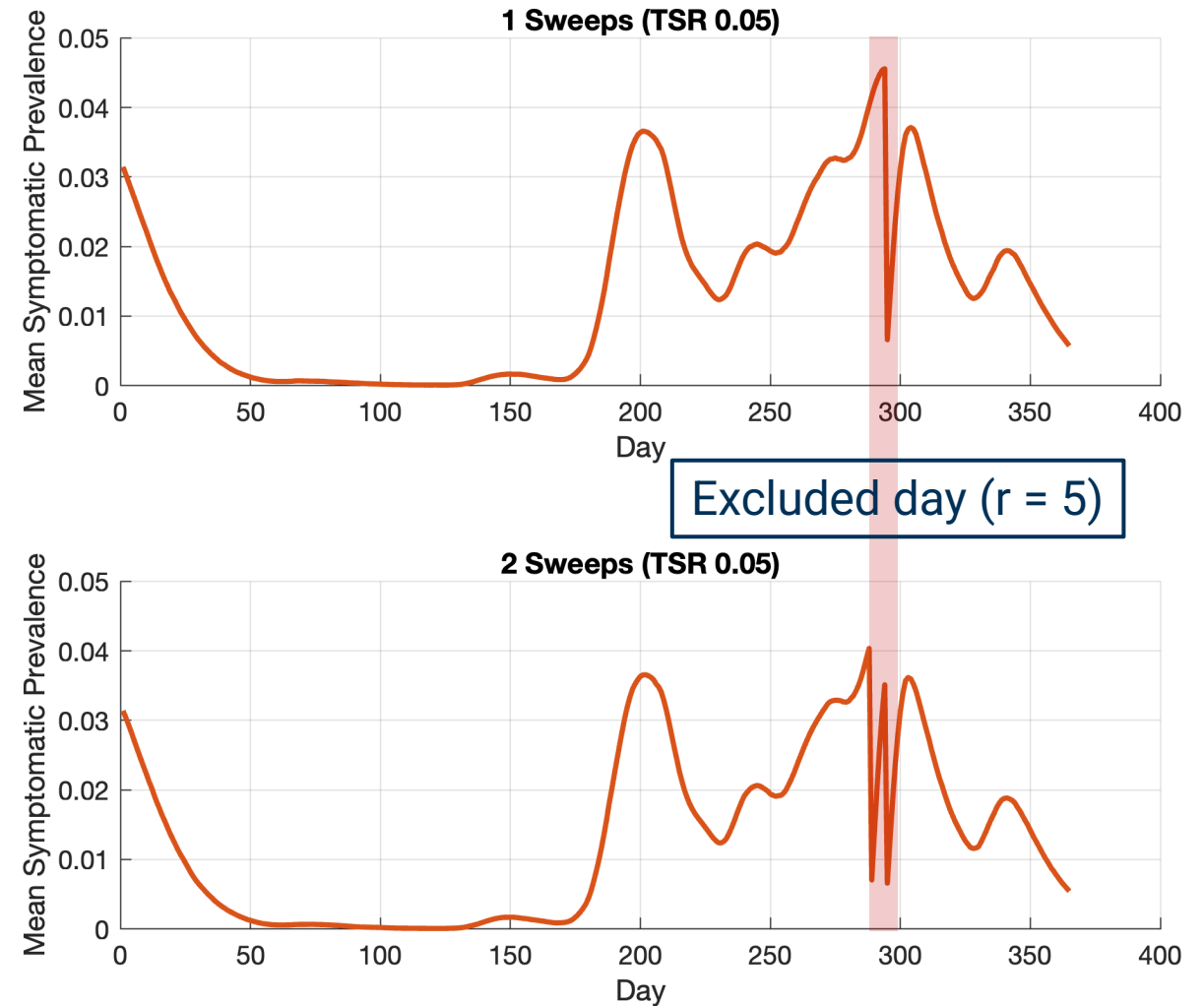
Uniform Strategy

- 20 weekly sweep in pilot study [1], fixed as planning horizon
- Generalized to uniform strategy
- Evenly space sweep days across the study period
- Easy to implement, for baseline comparison with other strategies

Greedy Strategy

■ Greedy Strategy

- Select sweep days iteratively
- Run simulation without sweeps
- Pick day with highest symptomatic prevalence as sweep day
- Keep selected day and rerun simulation
- Exclude nearby days and select the highest prevalence day and continue



Greedy Strategy

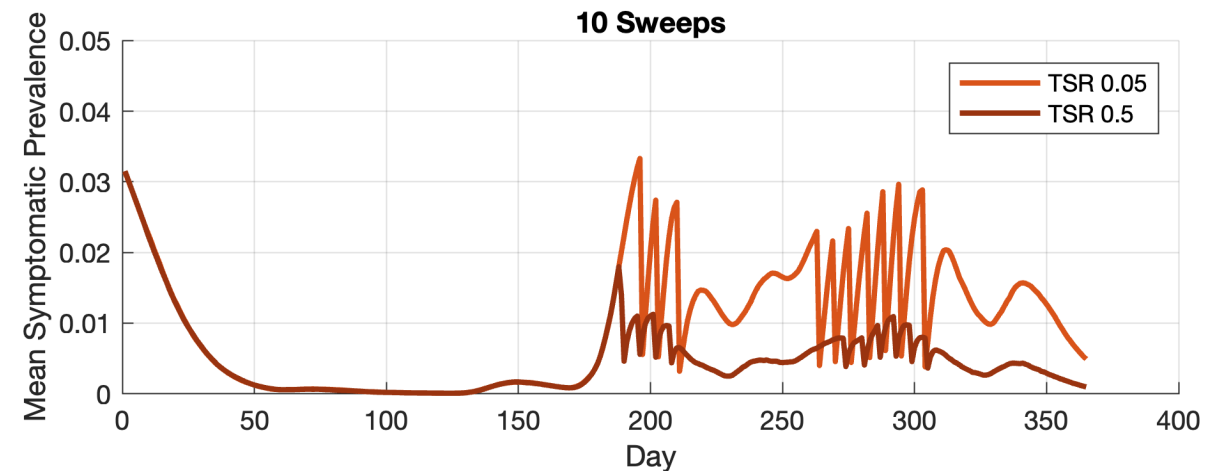
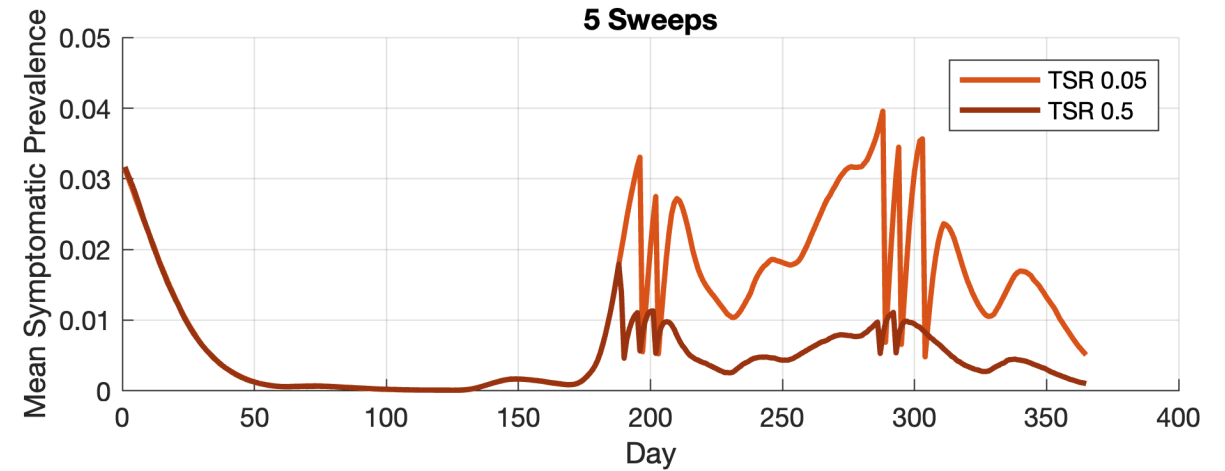
■ Greedy Behaviors (r=5)

- Concentrate sweeps on peaks
- Pattern stable to TSR



Effect of Different TSR

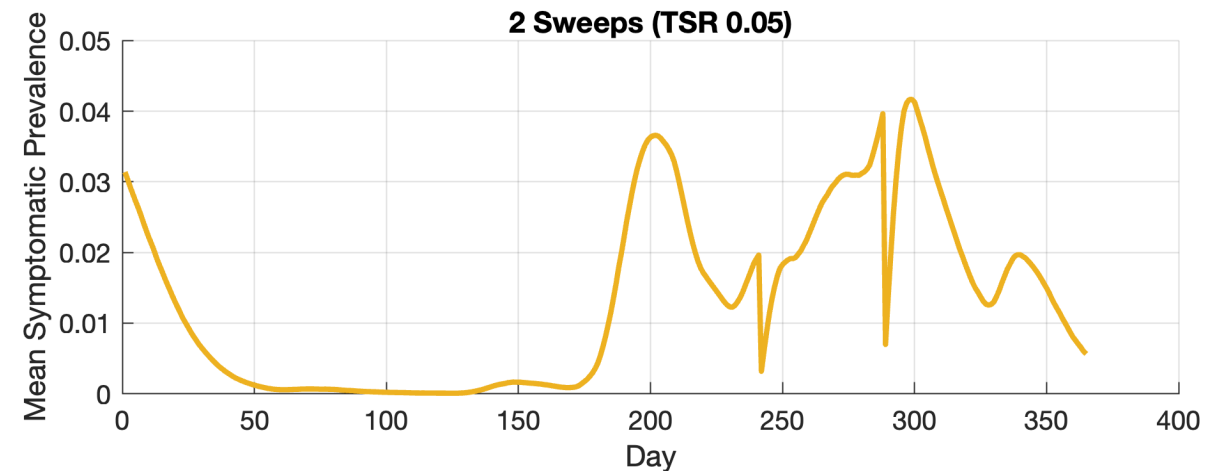
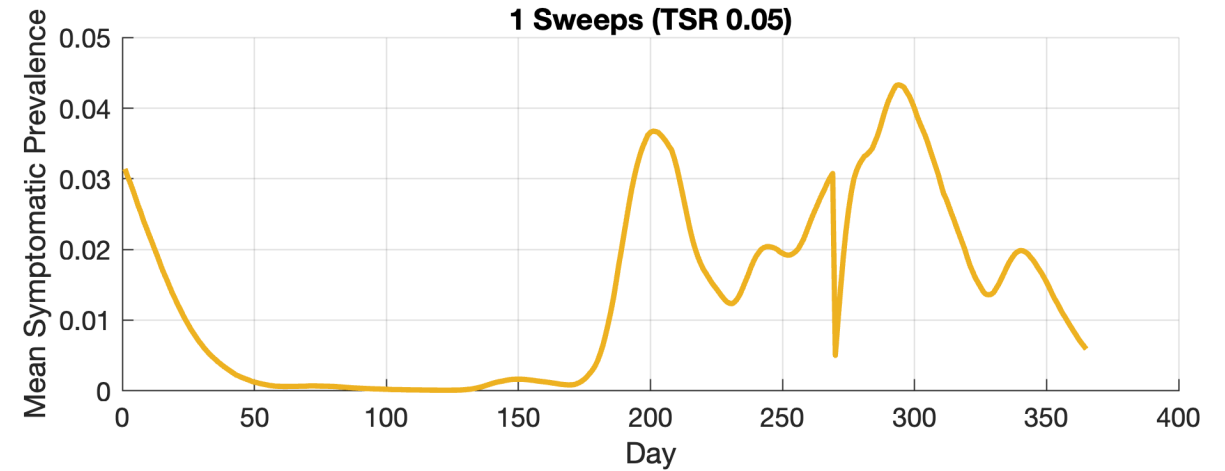
- Affect overall level of symptomatic prevalence
- Can't change the transmission pattern due to asymptomatic group



Weighted Strategy

■ Weighted Strategy

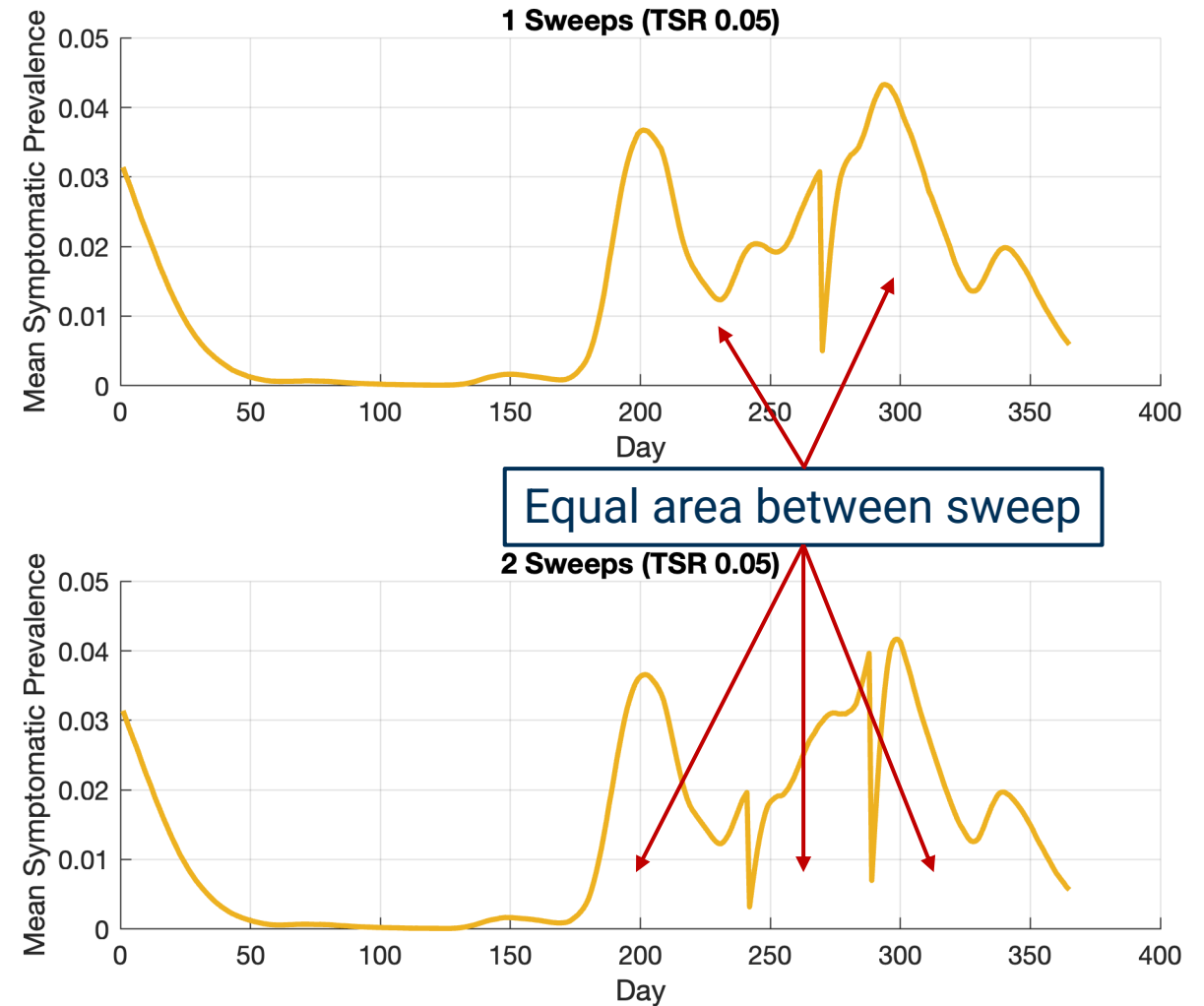
- Sweeps to balance disease burden
- Simulate baseline (no sweeps) to get daily symptomatic prevalence
- Compute cumulative prevalence over time and split evenly
- Assign split points as sweep days



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Weighted Strategy

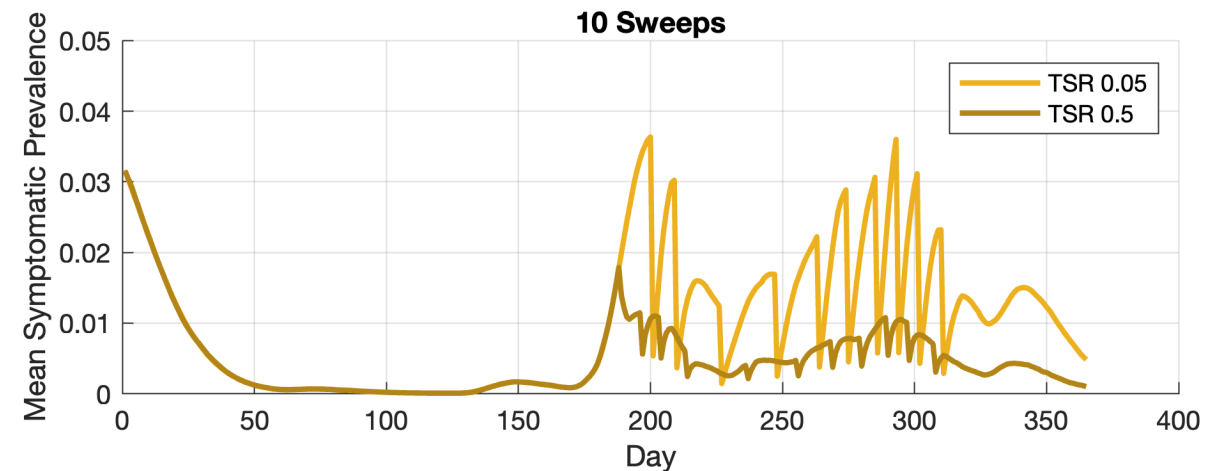
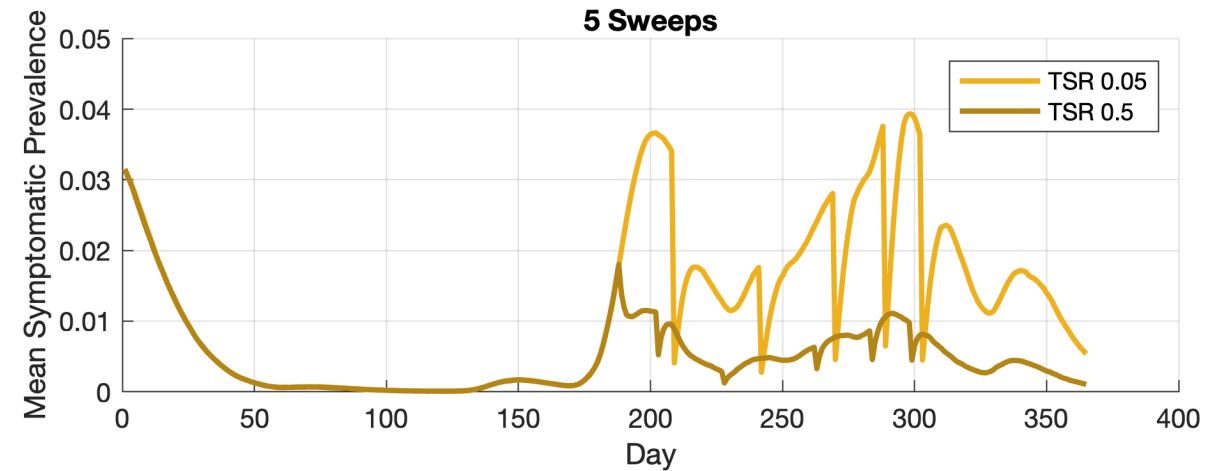
Weighted Behaviors

- Support temporal equity
- Pattern unstable to TSR



Effect of Different TSR

- Higher TSR flattens the curve
- Alter daily weight
- Shift sweep days in weighted strategy



Severe Malaria Cases



Performance Comparison

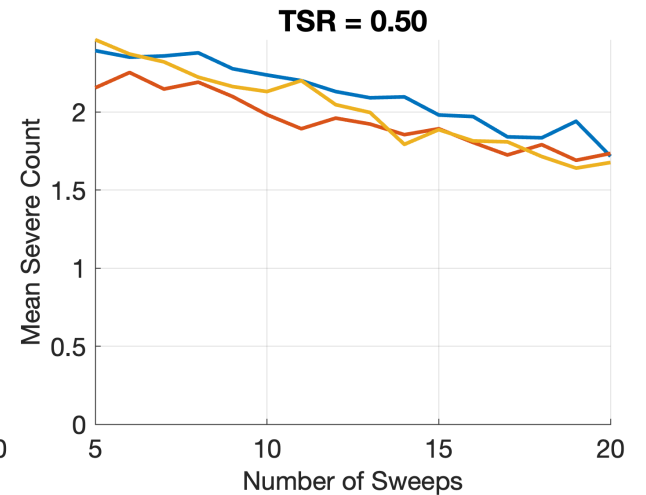
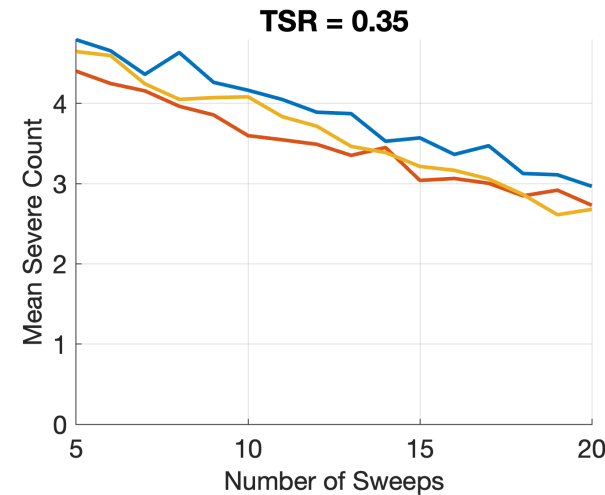
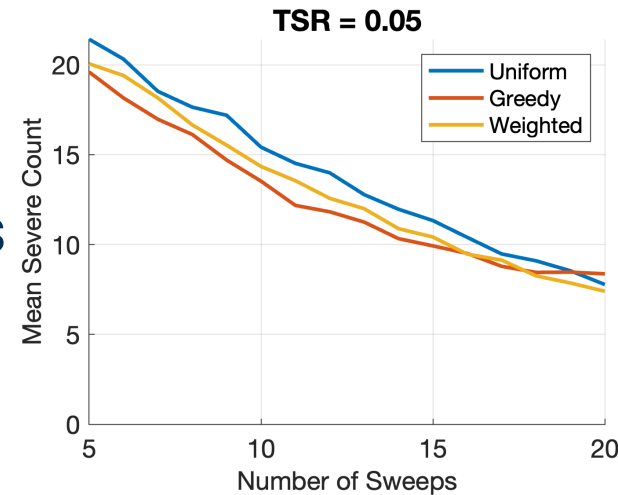
- Severe cases drop with more sweeps
- Greedy and Weighted outperform Uniform

- Biggest gain at low TSR



Effect of TSR

- Difference among strategies narrows
- High TSR reduces benefits



Conclusion & Future Work

✓ Conclusion

This study shows the value of data-informed sweep scheduling

- Resource efficiency is essential, especially under budget constraints
- Planning should reflect local natural and contextual characteristics
- Potential of optimization in resource allocation for disease control

🕒 Future Work

Toward more comprehensive, optimized intervention planning

- Broader interventions
- Simulation based optimization
- Under practical constraints
 - 💰 Limited budget
 - 👩 Local healthcare workers

Thanks !

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